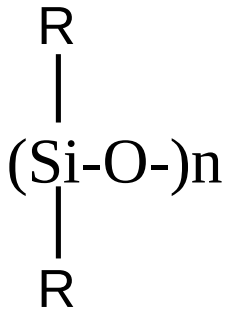
Welcome to Fascinating Silicone Chemistry

Silicone is not a product, but an entire field of chemistry

What is silicone?

The name "silicone" was given in 1901 by Kipping. Silicone is a generic name for a wide variety of polymeric chains and networks constructed around a backbone of Si-O-Si.

Technically known as "polysiloxanes," **silicones are the "missing link"** between organic and inorganic chemistry.




ORGANIC
Silicone
INORGANIC

Silicone vs. Organic polymers

Carbon	Silicon
C is organic (plant-based)	Si is inorganic (mineral-based)
C forms covalent bonds	Si forms hybrid ionic/covalent bonds
CO_2 is a simple gas	SiO_2 is a complex solid polymer
CCl_4 is a reasonably stable fluid	$SiCl_4$ is highly reactive to water and some organic substances

Bond energy – kJ/mole

C-C-C	C-O-C	Si-O-Si
~346	~358	~445

Bond length – angstroms (Å)

C-C-C	C-O-C	Si-O-Si
1.5	1.4	1.6

Barrier to rotation – kJ/mole

X	C-X	Si-X
– CH_3 (Me)	15.1	6.7
– O –	11.3	<0.8

RI
(Si - O -)n
IR

Some important differences between silicon and carbon

- The silicon atom is larger than the carbon atom. Its bonds are longer and more flexible with wider bond angles.
- Silicon is less electronegative than carbon (1.8 for silicon vs. 2.5 for carbon); it is able to give up more of its electrons to form strong, energetic bonds with other elements.
- Silicon only forms single bonds, not multiple bonds (single bonds are more stable and harder to break than double or triple bonds).

The Si-O bond – the key to silicone's unique properties

- The Si-O bond has higher bond energy than the C-O bond.

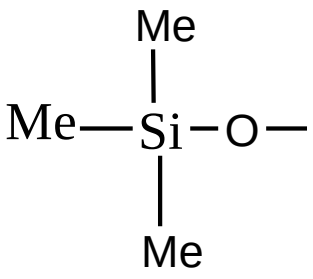
- The Si-O bond is longer and flatter than the
- C-O bond.

- The Si-O bond has a lower barrier to rotation than the C-O bond and higher free volume.

Rochow Process :

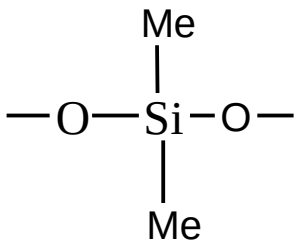
- **$Si + 2\ CH_3Cl$ ---------> $(CH_3)_2$-Si-Cl_2 (Predominate, 80%) + $(CH_3)_3$-Si-Cl, 4%)+ CH_3-Si-Cl_3, 12%) +**

  $Si-Cl_4$ + CH_3HSiCl_2, 3% + $(CH_3)_2HSiCl$+ Others



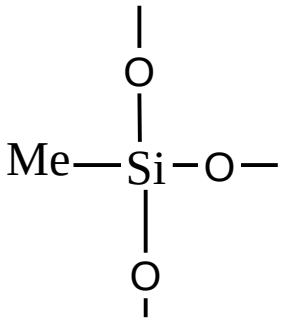
$Me_3SiO_{1/2}$

M



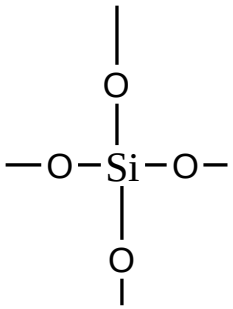
$Me_2SiO_{2/2}$

D



$MeSiO_{3/2}$

T



$SiO_{4/2}$

Q

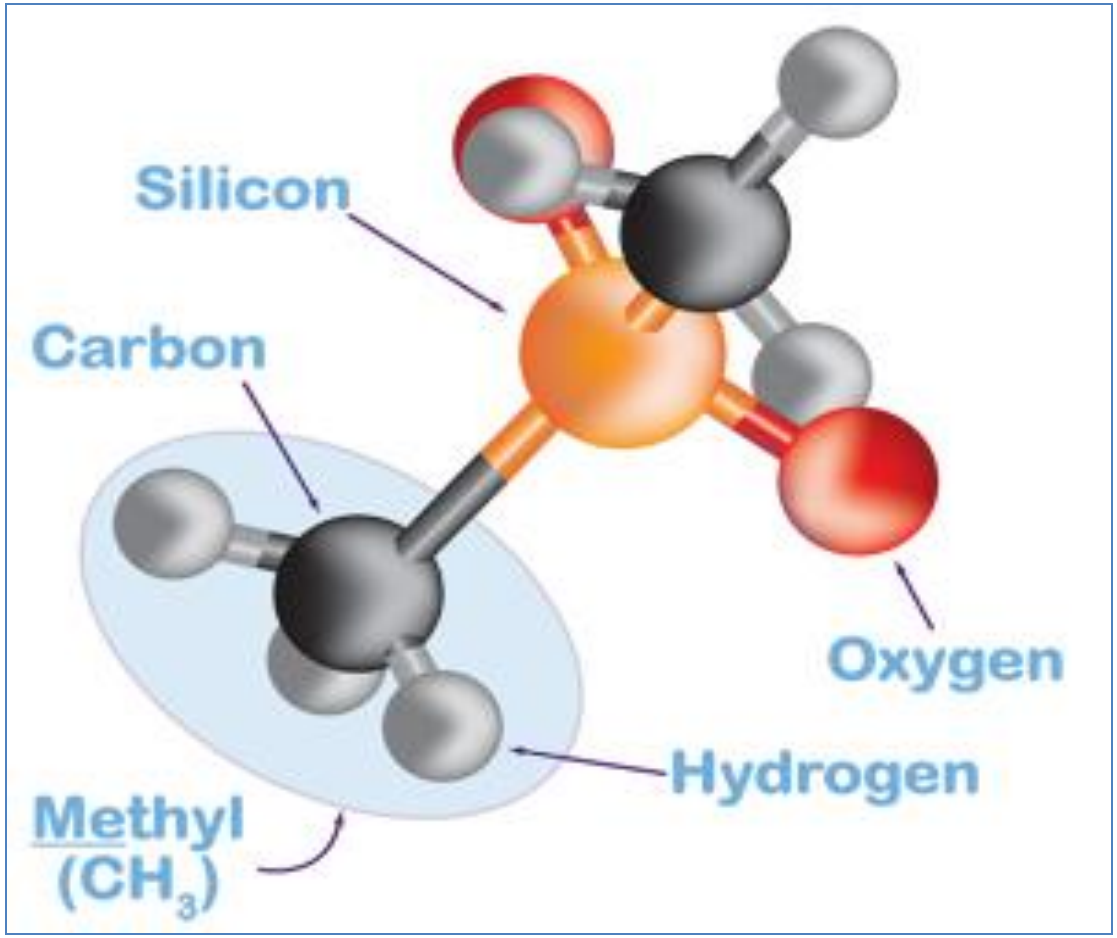
"M unit" is monosubstituted (one oxygen atom on silicon)

"D unit" is disubstituted (two oxygen atoms on silicon)

"T unit" is trisubstituted (three oxygen atoms on silicon)

"Q unit" is tetrasubstituted (four oxygen atoms on silicon)

Silicone building blocks



Preparation: From Sand to Silicones

- ***Silicone polymers. Silicone polymers are obtained by a three-step synthesis:***

1. Chlorosilane synthesis
2. Chlorosilane hydrolysis
3. Polymerisation and polycondensation

1. Chlorosilane synthesis.

- $SiO_2 + 2C \xrightarrow{1700^\circ C} Si + 2CO$

$$CH_3OH + HCl \xrightarrow{Catalyst} CH_3Cl + H_2O$$

Rochow Process :

- $Si + 2\ CH_3Cl \xrightarrow{Cu\ Catalyst} (CH_3)_2\text{-}Si\text{-}Cl_2$ (Predominate,
- 80%) + $(CH_3)_3\text{-}Si\text{-}Cl$, 4%)+ $CH_3\text{-}Si\text{-}Cl_3$, 12%) +

$Si\text{-}Cl_4 + CH_3HSiCl_2$, 3% + $(CH_3)_2HSiCl$+ Others

2. Hydrolysis of chlorosilane (to produce HCl and siloxanediol)

$(CH_{3)2}SiCl_2 + H_20$ ---> **$HCl + (CH_3)_2Si(OH)_2$**

Dehydration of siloxanediol to cyclomethicone and silanols

$x(CH_3)_2Si(OH)_2$ --> H_2O + y HO-$(CH_3)_2SiO)_nH$ + z {$(CH_3)_2SiO)m$} cylclomethicone

Linear	Cyclic
n = 20-50	m = 3, 4, 5 mainly 4

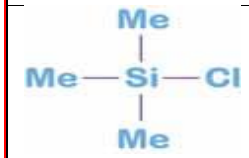
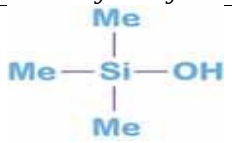
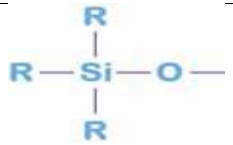
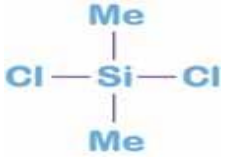
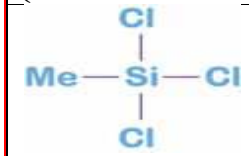
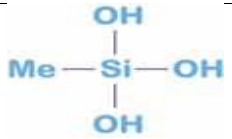
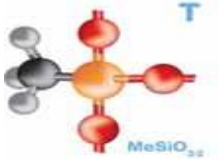
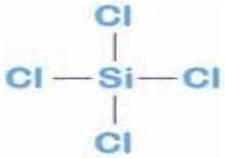
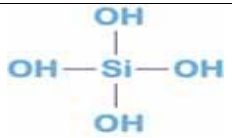
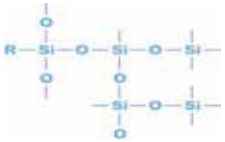
$(CH_3)_2Si(OH)_2$ condenses, with HCl acting as a catalyst, to give a mixture of linear or cyclic oligomers by inter- or intramolecular condensation.

3. Polymerisation and polycondensation

The linear and cyclic oligomers obtained by hydrolysis of the dimethyldichlorosilane have too short a chain for most applications. They must be condensed (linears) or polymerised (cyclics) to give macromolecules of sufficient length.

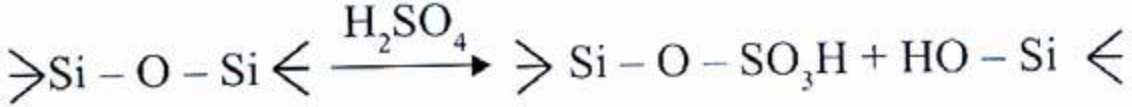
Cyclics, $(R_2SiO)_m$, can be opened and polymerised to form long linear chains

$$x\ (Me_2SiO)_4 + KOH \longrightarrow (Me_2SiO)_y + KO(Me_2SiO)_zH$$

Chlorosilane Precursor	Product of Hydrolysis	Silicone "Shorthand"	Forms, Functions
 Trimethylchlorosilane			 End-blockers
 **Dimethyldichlorosilane** The primary building block used in the polymerization of polydimethylsiloxanes (classic linear silicones)			Linear (straight chain)  Silicone fluids, elastomers, and gels
 Methyltrichlorosilane			Branched (crosslinkers) 
 Tetrachlorosilane			Three-dimensional resinous networks  Silicone resins

Silicone oil

- A **silicone oil** is any liquid polymerized siloxane with organic side chains. They are formed with a backbone of alternating silicon-oxygen atoms (...Si-O-Si-O-Si...) – i.e. siloxane – rather than carbon atoms (...
- A typical example is Hexamethyldisiloxane, where three methyl groups attach to each silicon atom to form $Si(CH_3)_3OSi(CH_3)_3$. The carbon analogue would be an alkylalkane, that is 2,2,4,4-tetramethylpentane $C(CH_3)_3CH_2C(CH_3)_3$.
- C-C-C-C)



Applications

- used as lubricants or hydraulic fluids,
- electrical insulators, freeze-dryers as refrigerants, anti-foaming agents
- Silicone oil is also one of the two main ingredients in **Silly Putty**

Silicone grease

- **Silicone grease** is a waterproof grease made by combining a silicone oil with a thickener. Most commonly, the silicone oil is polydimethylsiloxane and the thickener is amorphous fumed silica.

Application

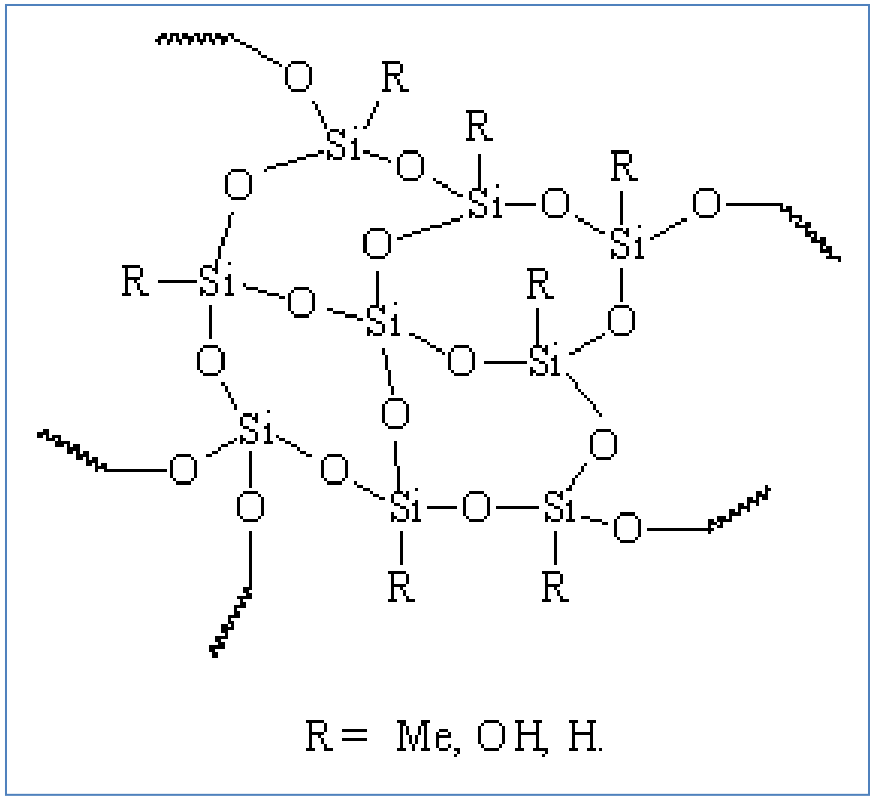
- lubricating and preserving rubber parts, such as O-rings
- Silicone grease is widely used as a temporary sealant and a lubricant for interconnecting ground glass joints, as is typically used in laboratory glassware
- Silicone-based lubricants are often used by consumers in applications where other common consumer lubricants, such as petroleum jelly,
- Used as dielectric grease in electrical connectors,

Silicone rubber

- **Silicone rubber** is an elastomer (rubber-like material) composed of silicone—itself a polymer—containing silicon together with carbon, hydrogen, and oxygen

Silicone resins

- **Silicone resins** are a type of silicone material which is formed by branched, cage-like oligosiloxanes with the general formula of $R_nSiX_mO_y$, where R is a non reactive substituent, usually Me or Ph, and X is a functional group H, OH, Cl or OR. These groups are further condensed in many applications, to give highly crosslinked, insoluble polysiloxane networks.[1]
- When R is methyl, the four possible functional siloxane monomeric units are described as follows:[2]
- "M" stands for Me_3SiO,
- "D" for Me_2SiO_2,
- "T" for $MeSiO_3$ and
- "Q" for SiO_4.
- The most abundant silicone resins are built of D and T units (DT resins) or from M and Q units (MQ resins), however many other combinations (MDT, MTQ, QDT) are also used in industry.


R = Me, OH, H.

Application

- Silicone resins find their application in many fields:
- *Silicone RTV-2 formulators*
- Vinyl functional resins for addition cured RTV-2 formulation
- Hydrogen functional resins for addition cured RTV-2 formulation
- *Personal Care*
- Trimethoxysiloxysilicate resins for film forming
- *Coating application*
- Binders
- Heat resistance
- *Construction*
- Façade treatment
- Water repellency

- **(i) Silicone oils: They have extremely high viscosity (*ca. 50,000 to 3,00,000 times that of water depending upon the chain length) which does not change appreciably with temperature. Their surface tension is low. Chemical inertness and presence non-polar groups make them non-toxic. They are used:***
- • As dielectric insulating media e.g. in transformer oils.
- • As hydraulic oils
- • As compressible fluids for liquid springs
- • As antifoaming agents in sewage disposal plants, textile dyeing, cooking oil etc. (ppm quantities do wonder!)
- • As an additive in cosmetics (lipsticks, sun tan lotion, hair oil etc.)

- **(ii) Silicone greases: Silicone oils are thickened to make silicone greases for use in heavy-duty steel gears and shafts. For this purpose, methyl phenyl silicone oils are thickened by Li-soaps.**

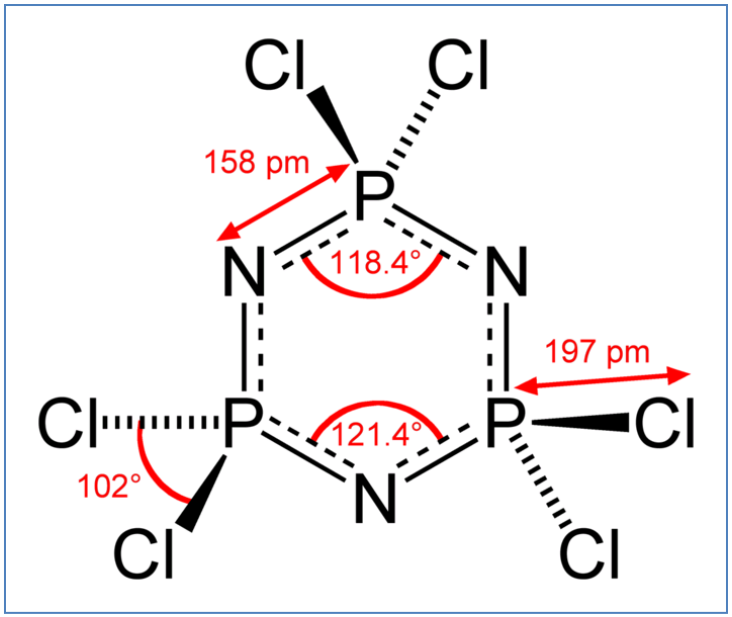
- **(iii) Silicone rubber: Silicon rubbers are used in making**
- • Cable insulation
- • Static and rotary seals
- • Gaskets
- • Diaphragms
- • Electric tape insulation
- • Industrial sealants and adhesives
- • Heart valves
- • Space suits
- • Accurate impressions for dentures
- • Masks in cinema.

- **(iv)Silicone resins: Silicone resins are used in making**
- • Insulations in electrical equipments
- • Laminates and printed circuit boards for electronic equipments
- • High temperature paints and coatings on cooking utensils.

Phosphonitrilic Compound

- **Phosphazenes** are a class of chemical compounds in which a phosphorus atom is covalently linked to a nitrogen atom by a double bond and to three other atoms or radicals by single bonds.

(phosphonitrilic chloride trimer)



Phosphonitrilic chloride,

$$PCl_5 + NH_4Cl \rightarrow 1/n\ (NPCl_2)_n + 4\ HCl$$

(phosphonitrilic chloride trimer)

Synthetic Routes to Polyphosphazenes . . . . . .

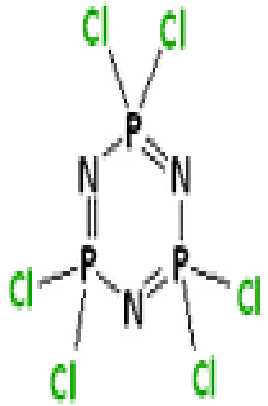
Method A
Substitution Reactions of Poly(dichlorophosphazene), $[Cl_2PN]_n$

Method B
Ring-Opening of Substituted Cyclic Phosphazenes . . . . . .

Method C
Condensation Polymerization of N-Silylphosphoranimines . . .

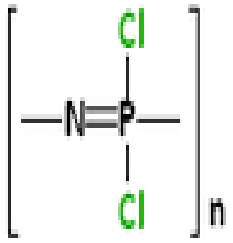
Method D
Derivatization of Preformed Poly(alkyl/arylphosphazenes) . . .

Step 1



Hexachlorocyclotriphosphazene

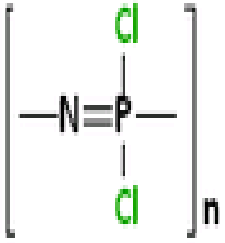
Heat



n = 15,000

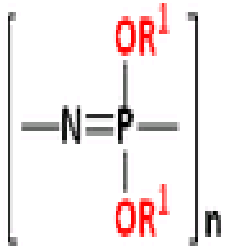
Poly(dichlorophosphazene)
(reactive polymer intermediate)

Step 2

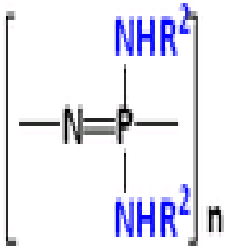


$RONa$, RNH_2, or R_2NH

· NaCl or HCl



or



or

